

Leader election – ring

```
#include <mpi.h>
#include <stdio.h>
#include <stdlib.h>
int token[5];
void show();
int main(int argc, char* argv[])
{

    MPI_Init(&argc,&argv);
    int rank,size;
    int uid[5]={111,222,333,444,555};
    int leader;

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (rank != 0)
    {
        MPI_Recv(&token, size, MPI_INT, rank - 1, 0, MPI_COMM_WORLD,
MPI_STATUS_IGNORE);
        printf("Process %d received token from process %d\n", rank,rank - 1);

        show();
        token[rank]=uid[rank];
    }

    else
    {
        token[rank] = uid[rank];
    }

    MPI_Send(&token, size, MPI_INT, (rank + 1) % size, 0,MPI_COMM_WORLD);

    if(rank!=0)
    {
        MPI_Recv(&leader,1,MPI_INT,(rank-1),1,MPI_COMM_WORLD,MPI_STATUS_IGNORE);
        printf("Process %d received token from process %d, Leader is %d\n", rank,rank - 1,
leader);
        MPI_Send(&leader,1,MPI_INT,(rank + 1)%size,1,MPI_COMM_WORLD);
    }

    if (rank == 0)
    {
        int i;
        MPI_Recv(&token, size, MPI_INT, size - 1, 0,
MPI_COMM_WORLD,MPI_STATUS_IGNORE);
        printf("Process %d received token from process %d\n", rank,size - 1);
```

```

show();

int max=token[0];
for(i=0;i<size;i++)
{
    if(max<token[i])
    {
        leader=i;
        max=token[i];
    }
}

MPI_Send(&leader,1,MPI_INT,(rank +1)%size,1,MPI_COMM_WORLD);
MPI_Recv(&leader,1,MPI_INT,(size-1),1,MPI_COMM_WORLD,MPI_STATUS_IGNORE);
printf("Process %d received token from process %d,leader is %d\n", rank,size-
1,leader);

}
MPI_Finalize();
return 0;
}

void show()
{
    int i;
    for(i=0;i<5;i++)
        printf("\n%d",token[i]);
}

```

Output

```

mpicc elect2.c -o e
mpirun -np 5 e

```

Process 1 received token from process 0

```

111
0
0
0

```

Process 2 received token from process 1

```

111
222
0
0

```

Process 3 received token from process 2

```

111
222

```

333

0

Process 4 received token from process 3

111

222

333

444

Process 0 received token from process 4

111

222

333

444

0Process 1 received token from process 0, Leader is 4

0Process 2 received token from process 1, Leader is 4

0Process 3 received token from process 2, Leader is 4

0Process 4 received token from process 3, Leader is 4

555Process 0 received token from process 4,leader is 4